

TradePilot to Cloud

A 4-week migration from a traveling laptop to a SEBI-ready, real-money-capable cloud system. Researched 2026-05-08 in response to live trading concerns.

AUTHOR

Soumya Swain

PROJECT

TradePilot · Mode A

DATE

2026-05-08 · v1.0

About the Author

Soumya Swain

Co-founder, Sidewall · IIM MBA · fintech operator

Email: soumya@sidewall.in

LinkedIn: linkedin.com/in/soumya-swain-iim

TradePilot is built and operated by Soumya as a personal R&D project, with paper-trading validation against Indian NSE markets.



LinkedIn

About this report

This document is the synthesis of four parallel research workstreams conducted on 2026-05-08 in response to a sequence of laptop-environment bugs that surfaced through the week of 2026-05-04. The full underlying research (provider comparison, cloud architecture, security and compliance threat model, and migration plan) lives alongside this PDF in the same folder. Cross-references are linked at the back of this document.

Copyright & Disclaimer

Project	TradePilot
Version	v1.0
Status	Active · awaiting Friday pre-flight
Created	2026-05-08
Updated	2026-05-08
Author	Soumya Swain

Disclaimer

TradePilot is currently in **Mode A — paper trading only**. All trades referenced in this document are simulated against real-time market data. No real capital is deployed. Cost projections, regulatory commentary, and provider recommendations are based on publicly available information as of 2026-05-08 and are not investment, legal, or tax advice. SEBI compliance requirements may change; always verify current rules with a qualified compliance advisor before deploying real money.

Research conducted by automated agents based on publicly available pricing pages, official documentation, and regulatory references. Pricing in INR uses an indicative USD-INR rate of 83 and may differ from actual provider invoicing. Treat all numbers as planning estimates, not contractual quotes.

This document is a private working artifact. Do not redistribute outside the TradePilot project without consent.

Contents

•	Executive Summary	7
1	Provider Comparison	9
2	Architecture	11
3	Migration Timeline	13
4	Security & Compliance	15
5	Cost Projection	17
6	Pre-Flight Checklist	18
7	The Decision	19

Page numbers are approximate; the document flows naturally and chapter headings appear at consistent positions.

Executive Summary

Trigger	Owner travels frequently · laptop in backpack with lid closed during market hours · 5 distinct laptop-environment bugs in past 2 weeks
Decision needed	Move TradePilot from laptop to cloud — by 2026-05-11 (this weekend) or 2026-05-18 (next weekend) at the latest
Recommendation	Phased migration: Render free tier → AWS Lightsail Mumbai → SEBI-compliant real money
Total cost (12 mo)	Rs 19,800 — Rs 35,000 depending on path
SEBI deadline	Static IP whitelisting with Zerodha required before any real-money trading

The 30-second version

The laptop was never going to be a sustainable platform for an algorithmic trading system run by someone who travels. Cloud isn't an upgrade — it's the only path that addresses the actual constraints: stay-awake-in-backpack, battery, thermal, network, and the SEBI static-IP requirement that's mandatory for real-money Kite Connect trading. The plan below moves to cloud in 3 phases, each one reversible, with the first phase done this weekend.

Three phases at a glance

Phase	Window	Goal	Cost	Risk
1 · Dashboard cloud	2026-05-10 / 11	Move Flask + paper-trading dashboard to Render free tier · laptop still runs engines · validates cloud setup	Rs 0/mo	LOW
2 · Engine cloud	2026-05-17 / 24	Move 7 engines + Rust to AWS Lightsail Mumbai (3GB) · static IP · laptop becomes backup	Rs 1,650/mo	MEDIUM
3 · Real money + security	2026-05-25 / 06-08	Kite Connect + JWT auth + audit logs + SEBI registration · paper-to-real-money cutover	+Rs 500-1,000/mo	HIGH (gated)

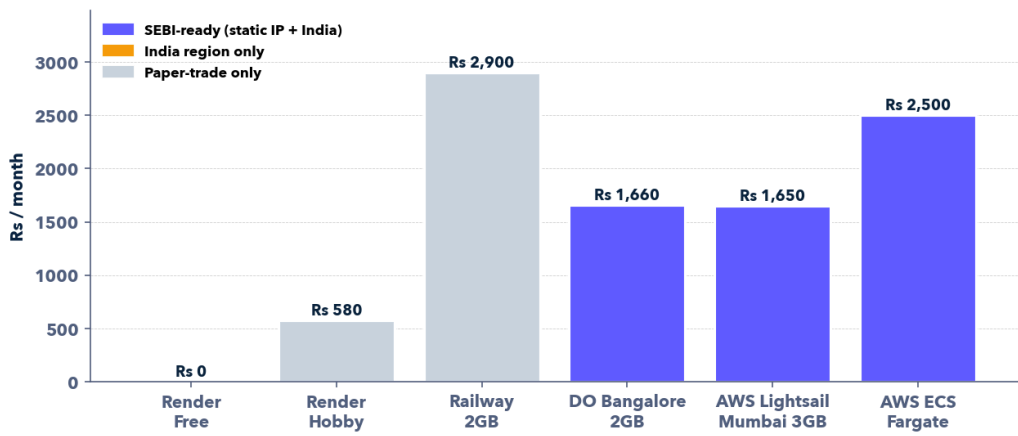
WHY PHASED

Each phase is reversible inside 10 minutes. Phase 1 doesn't risk real money (paper only). Phase 2 builds confidence with no money at stake. Phase 3 is the only phase with regulatory and financial risk — and it's gated by phases 1 and 2 succeeding.

1 · Provider Comparison

Four agents researched 10+ providers across pricing, India region availability, static-IP support, and workload fit. The cost picture below shows what matters: **SEBI-ready (real-money) deployment requires a static IP in India**. PaaS-style services like Render and Railway have rotating IPs and are paper-trade-only.

Monthly cost by provider



The shortlist

Provider	Plan	RAM	India	Static IP	Rs/mo	Best for
Render	Free	512MB	No	No	0	Phase 1 — dashboard only
Render	Hobby	2GB	No	No	580	Paper-trade scaling
Railway	Hobby	2GB	SG (~150ms)	No	2,900	Skip — high latency
DigitalOcean	Bangalore 2GB	2GB	Yes	Yes (paid)	1,660	Phase 2 alternative
AWS Lightsail	Mumbai 3GB	3GB	Yes	Yes	1,650	Phase 2 winner
AWS ECS Fargate	Auto-scaled	Variable	Yes	Yes (NAT)	2,500+	Future scale (Phase 4+)

PHASE 2 WINNER: AWS LIGHTSAIL MUMBAI 3GB AT RS 1,650/MONTH

Sub-10ms latency to NSE servers · static IP that satisfies Kite Connect's whitelisting requirement · simpler than ECS Fargate for a one-person team · auto-renewing SSL · Ubuntu 22.04 with full SSH access · India region keeps you compliant with data residency.

2 • Architecture

Current architecture: `launchd` + `nohup` + `pgrep` on macOS. Cloud target: **Docker Compose on AWS Lightsail** (Phase 2) with optional ECS Fargate path for Phase 4+ if scale demands it. Kubernetes is overkill for 7 background workers and 1 web app.

Service breakdown

Service	Type	RAM	Restart policy
Flask dashboard	Web (Gunicorn 2 workers)	~250 MB	always
Rust engine	HTTP service (port 8081)	~100 MB	always
v4 paper-trade engine	Long-running script	~200 MB	on-failure
v5 / v5_classic / v5_6 / v5_7 / v5_8 / v6	6 long-running scripts	~150 MB each	on-failure
EOD report cron	Scheduled (16:11 IST)	~100 MB	cron
Telegram digest cron	Scheduled (every 30 min)	~50 MB	cron

State persistence – keep it simple

Current: ~50 MB of JSON files in `prototype/data/` and `prototype/v*/`. Cloud option A (recommended): **persistent volume** on Lightsail, JSON files keep working. No DB migration. Cloud option B (later, when Phase 4):

PostgreSQL for trade history, S3 for ML weights. **Don't migrate to DB yet** — adds 2 weeks of work for marginal gain at this stage.

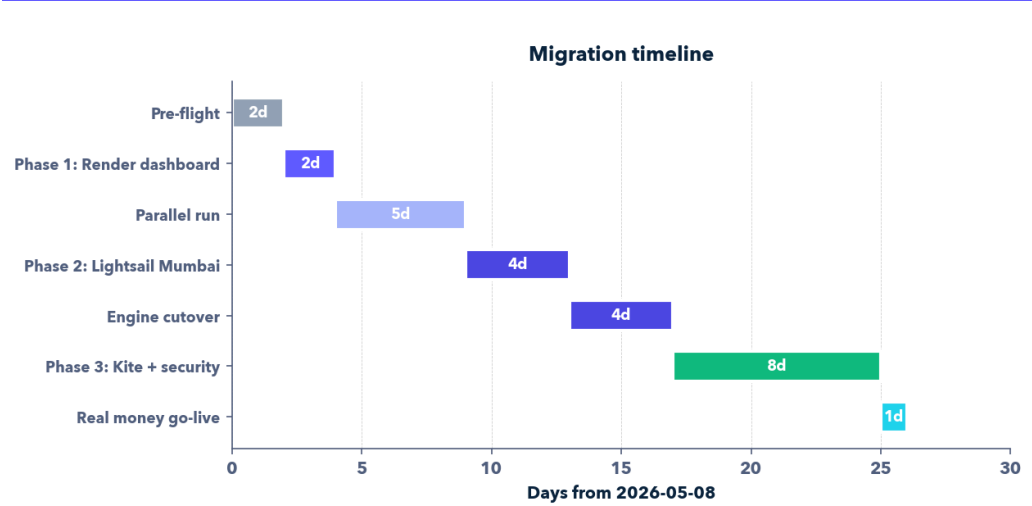
Process orchestration

Replace `launchd` with **Docker Compose with restart policies**. One `docker-compose.yml` with the 9 services above. `docker compose up -d` on Lightsail = full system running. Cron-style scheduled jobs (auto-stop-eod, EOD report, Telegram digest) handled by **system cron inside the container** — simpler than EventBridge for a one-person team.

CI/CD – start manual, automate later

Phase	Deploy method	Rollback
1 (Render)	git push, auto-deploy via Render webhook	Render dashboard "rollback to previous" button
2 (Lightsail)	git pull + docker compose restart on the VM (manual SSH)	git checkout previous SHA + restart
3+ (production)	GitHub Actions, SSH deploy with health-check gate + market-hours block	Automated rollback on health-check failure

3 • Migration Timeline



This weekend (Phase 1)

When	Task	Effort	Verifies
Fri 2026-05-09 evening	Render account · GitHub repo connected · review existing <code>render.yaml</code>	30 min	Cloud account works
Sat 2026-05-10 09:00	Pre-flight checks · Docker builds locally · <code>.env.cloud</code> drafted	1 hr	Code is cloud-portable
Sat 10:00 – 12:00	First deploy to Render free tier · dashboard reachable at <code>.onrender.com</code> URL	2 hrs	HTTP 200, HTTPS valid
Sat 12:00 – 15:35	Parallel run · cloud dashboard alongside laptop · compare <code>/api/engine-status</code> data	Watch only	Data parity
Sat 16:00	EOD review — cloud rendered correctly?	15 min	End-to-end works
Sun 2026-05-11	Run cloud dashboard read-only all day · take screenshots from phone	Watch only	Mobile-accessible
Sun 14:00	Go/no-go for Phase 2 next weekend	15 min	Confidence gate

Cutover gates – Phase 1 must pass all 6 before Phase 2 begins

1. Cloud dashboard returns HTTP 200 from any device on any network
2. HTTPS certificate is valid (browser shows lock icon)
3. `/api/engine-status` data matches laptop's data within 60 seconds
4. Logs visible in Render dashboard (or AWS CloudWatch)
5. No 500 errors in first 24 hours of running
6. Owner can interpret cloud logs without help

Phase 2 (next weekend)

2026-05-17 (Sat): Provision AWS Lightsail Mumbai 3GB. SSH access verified. Dockerfile multi-stage builds correctly. Static IP attached. **2026-05-18 (Sun):** Deploy 7 engines + Rust + cron via docker-compose. Run

alongside laptop with state files synced via S3. Compare BUY/HOLD counts and trade decisions side-by-side for 24 hours. Cutover Monday morning if parity holds.

Phase 3 (week of 2026-05-25)

Add JWT auth (Flask-JWT-Extended), Flask-Limiter (10 req/min), Doppler for secrets, Cloudflare in front of Lightsail. Apply for Kite Connect production app (Rs 2,000/month). Submit static IP to Zerodha for whitelisting. Begin 5-day shadow trading: signals computed on cloud, orders go to Kite paper account first, then real account with 1% capital, scaling up over a week.

4 • Security & Compliance

SEBI COMPLIANCE • THE NON-NEGOTIABLE

Individual algo traders are exempt from formal algo registration if trading **under 10 orders per second**. TradePilot targets 2-4 OPS, so we're inside the exemption. **But** Zerodha requires static IP whitelisting for any production-grade Kite Connect access — this means real money cannot trade from a laptop with a rotating IP, and not from Render or Railway either. **This single requirement determines the cloud provider choice.**

Threat model – the top 6

Threat	Likelihood	Impact	Mitigation
Kite API key leaked via git commit	Possible	Critical	Doppler secrets · pre-commit hook · GitHub secret scanning
Flask /api/paper/buy abused without auth	Likely	Major	JWT on all write endpoints · Flask-Limiter 10 req/min/IP
State file corrupted mid-write during outage	Possible	Major	Atomic writes (write-tmp-then-rename) · S3 hourly backups
Cache-poisoning style bug recurs	Likely	Moderate	Pre-market cache write block · 5-min TTL · all-NaN write rejection
Cloud provider outage during market	Unlikely	Major	Failover to laptop (paper only — real money pauses) · status-page subscription
Bad commit deploys during market hours	Possible	Critical	GitHub Actions: block production deploys 09:15-15:30 IST · canary deploy gate

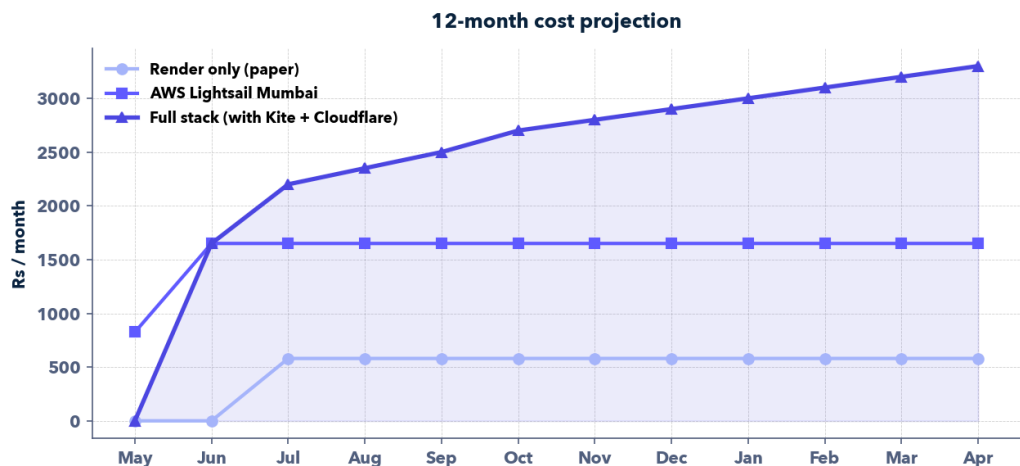


Critical pre-launch tasks (real-money gate)

1. **Static IP registered with Zerodha** — submit Lightsail IP via Kite Connect dashboard, wait 1-2 business days for whitelist confirmation

2. **JWT auth on all write endpoints** — Flask-JWT-Extended, 15-min access tokens, 7-day refresh tokens
3. **Doppler secrets management** — Kite api_key + api_secret + access_token (rotates daily) · Telegram bot token · Render/AWS credentials
4. **HTTPS-only enforcement** — Render auto-provisions, Lightsail needs Let's Encrypt via certbot
5. **Audit log to S3 Glacier** — every order decision logged immutably, 5-year retention per SEBI
6. **Rate limiting** — Flask-Limiter on all endpoints to prevent abuse and cost spikes
7. **Market-hours deploy block** — GitHub Actions check refuses production deploy between 09:15-15:30 IST
8. **Cloudflare in front** — DDoS protection · WAF · Rs 1,650/month Pro tier (only needed for Phase 3)

5 • Cost Projection



Where the money goes

Item	One-time	Monthly	Annual
Domain (tradepilot.in or similar)	Rs 950	—	Rs 950
Render (Phase 1, dashboard)	—	Rs 0 free / Rs 580 hobby	Rs 0 — Rs 6,960
AWS Lightsail Mumbai 3GB (Phase 2)	—	Rs 1,650	Rs 19,800
Kite Connect (Phase 3)	—	Rs 2,000	Rs 24,000
Cloudflare Pro (Phase 3, optional)	—	Rs 1,650	Rs 19,800
Doppler (free tier sufficient)	—	Rs 0	Rs 0
S3 backups + CloudWatch logs	—	Rs 250	Rs 3,000
Year 1 total · paper-trading only	Rs 950	Rs 1,900 avg	Rs 23,750
Year 1 total · with real money	Rs 950	Rs 5,550 avg	Rs 67,550

THE KITE CONNECT COST IS THE NEW LINE ITEM

Rs 2,000/month is real money's price of admission. It buys you a stable WebSocket tick stream (eliminates the entire yfinance NaN/cache/poll category of bugs) and the legal right to place real orders programmatically. Worth it the day you flip to real money — not before.

6 • Pre-Flight Checklist

Skills to learn before Saturday morning (~2 hours)

1. **Docker basics** — `docker build`, `docker run`, `docker ps`, `docker logs`. Practice on local machine first.
2. **Reading cloud logs** — Render dashboard → Logs tab. AWS Lightsail → SSH + `docker logs <service> -f`

3. **HTTPS & DNS basics** — what an A record does, what a CNAME does, why HTTPS matters
4. **SSH into a VM** — `ssh -i key.pem ubuntu@<ip>` · navigate, view logs, restart services
5. **Reading deployment errors** — common ones: missing env var, wrong Python version, port already in use

Pre-flight checklist (do by Friday EOD)

- Render account created · payment method added (even for free tier — required)
- GitHub repo public OR Render given access to private repo
- Local `docker build .` succeeds against current `Dockerfile`
- Local `docker compose up` brings up Flask + Rust + 7 engines
- Backup taken of `~/Desktop/TradePilot/` from yesterday — git push to origin too
- `.env.cloud` created (a copy of `.env` with Render-specific paths)
- Domain purchased (optional but recommended — `tradepilot.in` or similar at Rs 950)
- Reviewed this PDF end-to-end · understands the 3 phases · agrees to phase order

Saturday morning – go/no-go gate

If **any** of these are FALSE on Saturday 09:00, postpone Phase 1 to next weekend:

1. Local docker compose runs all 9 services without crashes
2. You slept >6 hours · not jet-lagged · not in transit
3. Stable internet · not on hotel Wi-Fi or mobile tether
4. You can dedicate the full Saturday — no client calls, no other commitments
5. Markets are closed (Saturday is fine for India)

Rollback runbook – if Phase 1 breaks Monday morning

Step	Action	Time
1	Render dashboard → Service settings → Suspend	30s
2	Confirm laptop engines still running: <code>./scripts/launch-market.sh --status</code>	30s
3	If laptop is also down: launch it: <code>./scripts/launch-market.sh</code>	1 min
4	If you bought a custom domain: revert DNS A record to laptop tunnel (or skip)	5 min
5	Telegram self-message: "Cloud rolled back, laptop is primary"	30s
Total rollback		<10 minutes

7 • The Decision

Option A • Cloud this weekend (recommended)

What: Render free tier · dashboard only · laptop continues running engines.

Risk: **LOW** — pure addition, nothing existing breaks.

Effort: 4-6 hours Saturday.

Outcome: Cloud dashboard reachable from anywhere, validates the migration concept, sets up Phase 2.

Option B • Cloud next weekend

What: Same as A but week of prep first.

Risk: **LOWER** — more time to learn, build confidence.

Effort: 1-2 hours/day this week + 4-6 hours next Saturday.

Outcome: Same as A, just a week later.

Option C • Skip Phase 1, jump to AWS Lightsail next weekend

What: Skip Render · go directly to Lightsail Mumbai.

Risk: **MEDIUM** — first cloud deploy is also your production deploy. No safety net.

Effort: 8-12 hours over the weekend.

Outcome: Static IP from day 1 · no Render-to-Lightsail re-migration later · costs start at Rs 1,650/mo immediately.

RECOMMENDATION: OPTION A

Phase 1 this weekend on Render free. It's the cheapest way to learn that cloud deployment works for your code without risk. Phase 2 next weekend on AWS Lightsail. By 2026-05-19 you have a SEBI-ready cloud system. By 2026-06-08 you can flip to real money. **Total elapsed: 31 days.** Total cost in those 31 days: Rs 950 (domain) + Rs 0 (Render free) + Rs 825 (half-month Lightsail). **About the price of one decent dinner.**

References

- [docs/research/2026-05-08_cloud_providers.md](#) — provider comparison detail
- [docs/research/2026-05-08_cloud_architecture.md](#) — service breakdown, deployment files
- [docs/research/2026-05-08_cloud_security.md](#) — SEBI, OWASP secrets, threat model
- [docs/research/2026-05-08_cloud_migration_plan.md](#) — day-by-day timeline
- [docs/planning/WEEKEND_PLAN_2026-05-10.md](#) — execution playbook

Decision time.

The laptop got us here.
It won't get us where you're going.

2026-05-10 · 09:00 IST

The first deploy.



soumya@sidewall.in